

MTH 446: Lecture # 13

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September 25, 2026

Lecture Span

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Uniqueness property

Definition 0.1. If $\{z \in D, f(z) = g(z)\}$, then $f = g$ for all $z \in D$. Note, f and g must be analytic functions on $\mathbb{C}$.

Example

Note that $\sin z$ and $\cos z$ are analytic in $\mathbb{C}$. Then, we wish to prove that

$$\cos^2 z + \sin^2 z = 1 \quad (1)$$

Since we know that $\cos^2 x + \sin^2 x = 1$ for all $x \in \mathbb{R}$. Therefore, since $\cos, \sin \in H(\mathbb{C})$, then it follows that

$$f(z) = \cos^2 z + \sin^2 z = g(z) = 1, \quad \forall z \in \mathbb{C} \quad (2)$$

Taylor's expansion theorem, if $f \in H(D_R(z_0))$, then

$$f(z) = \sum a_n(z - z_0)^n, \quad |z - z_0| < R \quad (3)$$

Note that such a function is infinitely differentiable. Analytic functions are relative to polynomials.

Theorem 0.2. Suppose $f'(z) = 0$ for all $z \in D$. Then, f is constant everywhere in D .

Proof. Let $f = u + iv$. Then, by the CR equations, we have

$$u_x = v_y, \quad u_y = -v_x \quad (4)$$

Therefore,

$$f'(z) = u_x + iv_x = 0 \quad (5)$$

Then

$$u_x \equiv 0 \equiv v_y \quad (6)$$

$$v_x \equiv 0 \equiv u_y \quad (7)$$

Therefore,

$$\nabla u \equiv 0 \implies u = \text{const.} \quad (8)$$

$$\nabla v \equiv 0 \implies v = \text{const.} \quad (9)$$

Therefore, f is constant throughout D . □

Some notes:

1. A point in which f is not analytic is a singular point.
2. A singular point z_0 of f is said to be isolated if $\exists \epsilon, f \in H(\overset{\circ}{D}_\epsilon(z_0))$.

Example

$$f(z) = \frac{1}{z^4}, \quad z \neq 0 \quad (10)$$

Note that $1 \in H(\mathbb{C})$ and $z^4 \in H(\mathbb{C})$. And $z^4 \neq 0$ if $z \neq 0$. Therefore, f is origin across $\mathbb{C}$ except for the origin. In other words,

$$f \in H(\mathbb{C}/\{0\}) \quad (11)$$

Therefore, 0 is an isolated singular point.

Example

$$f = \frac{2z + 1}{z(z^2 + 1)} \quad (12)$$

The denominator is zero if $z = 0, i, -i$. Therefore,

$$f \in H(\mathbb{C}/\{0, i, -i\}) \quad (13)$$

Therefore, $0, i, -i$ are isolated singular points. Note, if the set is finite, then all the values in the set are isolated singular points.

Example

$$f(z) = x^2 + y^2 \quad (14)$$

Both u and v are continuously differentiable on $\mathbb{C}$.

Previously, if $z \neq 0$, then f is not differentiable. However, since f' is not differentiable on a neighborhood, it follows that f' is not analytic at all.

Example

$$f = xy + iy \quad (15)$$

We have established that f' DNE if $z \neq 2$ and $f'(2) = 0$. Therefore, we can establish that f is not analytic at $z \neq 2$. Now, consider $z = 2$, no neighborhood. Therefore, f is nowhere analytic.

Proposition 0.3. f must be constant in D if one of the following is satisfied.

1. f is a real-valued function
2. $\bar{f} \in H(D)$
3. $|f(z)| = \text{const. in } D$

Proof. We first prove (a). Note that $f \in H(D)$. Also consider $f = u + i(0)$. Then,

$$u_x = v_y = 0 \iff u = \text{const.} \quad (16)$$

We can show that $\nabla u = 0$ and $\nabla v = 0$. Therefore, f is constant on D .

We next prove (b). We note that $f, \bar{f} \in H$. Note that we can rewrite v and u in terms of f and $\bar{f}$. But then, u and v are real valued. Therefore, by (a), we have that f is const. in D .

Finally, we prove (c). Note that $|f|^2 = c^2$. Also, $f \in D$. Firstly, if $c = 0$, then f is 0 and therefore is constant. Next, if $c \neq 0$, then note that f can never be 0 for all $z \in D$. Next, $\bar{f}f = c^2$. Then,

$$\bar{f} = \frac{c^2}{f} \in H(D) \quad (17)$$

Therefore, f is constant in D by (b). □

We introduce notation:

1. $c^1(D)$ is the set of all real valued functions u such that u, u_x, u_y exist and are cont. in D

2. $C^2(D)$ is the set of all real valued funcs u such that u, u_x, u_y, u_{xx} , etc. exist and are cont. in D .

Definition 0.4. 1. $C^0(D)$ is the space of continuous functions in D

2. $C^k(D)$ is the space of all functions u with continuous partial derivatives of order j .

3. $C^\infty(D)$ is the set of all infinitely differentiable functions in D .

Definition 0.5. A function u is called harmonic if $u \in C^2$ and $\nabla^2 u = 0$.

Notation: $\text{Har}(D)$ is the set of all harmonic functions in D .

Theorem 0.6. $f \in H(D)$ and $f = u + iv$. Then $u, v \in \text{Har}(D)$.